Mustafa Cavus, Ph.D.
Department of Statistics, Eskisehir Technical University
26555 Eskisehir, Türkiye
Email: mustafacavus@eskisehir.edu.tr | **Date:** November 9, 2026

To: The Editorial Board and Guest Editors
Special Collection on "Ethics of Digital Transformation in Tourism and Hospitality"
Information Technology & Tourism, Springer Nature

Subject: Manuscript Submission

Dear Editors,

I am pleased to submit my research manuscript entitled **"Explainable Hotel Booking Cancellation Prediction Models: Towards Trustworthy Survival Analysis in Hospitality"** for consideration in *Information Technology & Tourism* within the Special Collection on **"Ethics of Digital Transformation in Tourism and Hospitality"**.

Context and Importance of the Work
As the hospitality industry rapidly adopts automated algorithmic systems, hotels increasingly depend on machine learning models to anticipate booking cancellations and manage capacity. However, conventional research frames cancellations as a static binary classification problem. This approach creates two fundamental vulnerabilities: it ignores the continuous temporal progression of attrition risk, and it introduces systematic predictive bias by treating active, right-censored bookings as non-canceled. Furthermore, delegating overbooking decisions to uncalibrated, opaque black-box models frequently leads to guest displacement and acute consumer insecurity, creating serious operational and ethical challenges.

To address these issues, this study develops an explainable time-to-event (survival analysis) framework evaluated on a large-scale empirical dataset of resort and city hotel reservations with 119,390 reservations. By benchmarking ten survival algorithms with non-parametric bootstrap uncertainty estimation, the study uncovers an essential governance insight: while standard tree ensembles maximize discriminative ranking, Oblique Random Survival Forests provide superior probability calibration. Calibrated probabilities serve as an operational and ethical safeguard, preventing erratic overbooking and protecting vulnerable travelers. Additionally, we integrate continuous time-dependent explainable AI tools (SurvPFI and SurvPDP) to audit model predictions, revealing time-varying risk dynamics and exposing structural policy anomalies in non-refundable reservations.

Appropriateness for the Journal and Special Collection
This work directly aligns with the mission of *Information Technology & Tourism* and the thematic focus of this Special Collection. By connecting advanced survival modeling with responsible technology governance, the paper demonstrates how calibrated predictions and explainability-by-design operationalize ethical AI principles in digital tourism ecosystems. The framework equips hospitality managers and platform governors with auditable decision-support tools that mitigate consumer insecurity and strengthen algorithmic accountability.

This manuscript represents original work that has not been published previously and is not under consideration elsewhere. All data and reproducible experimental pipelines are openly accessible at: GitHub Repository.

Thank you for your time and consideration.

Sincerely,

Mustafa Cavus, Ph.D.
Eskisehir Technical University, Department of Statistics
Email: mustafacavus@eskisehir.edu.tr